

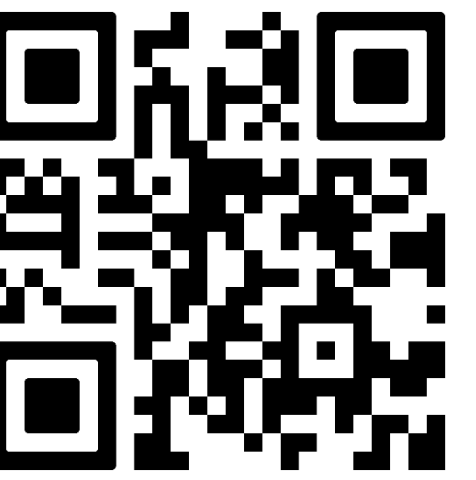
Robustness Enhancement for Multi-Quadrotor Centralized Transportation System via Online Tuning and Learning

Tianhua Gao, Kohji Tomita, Akiya Kamimura

For further information: gao.tianhua@ieee.org Corresponding: kamimura.a@aist.go.jp

Simulation

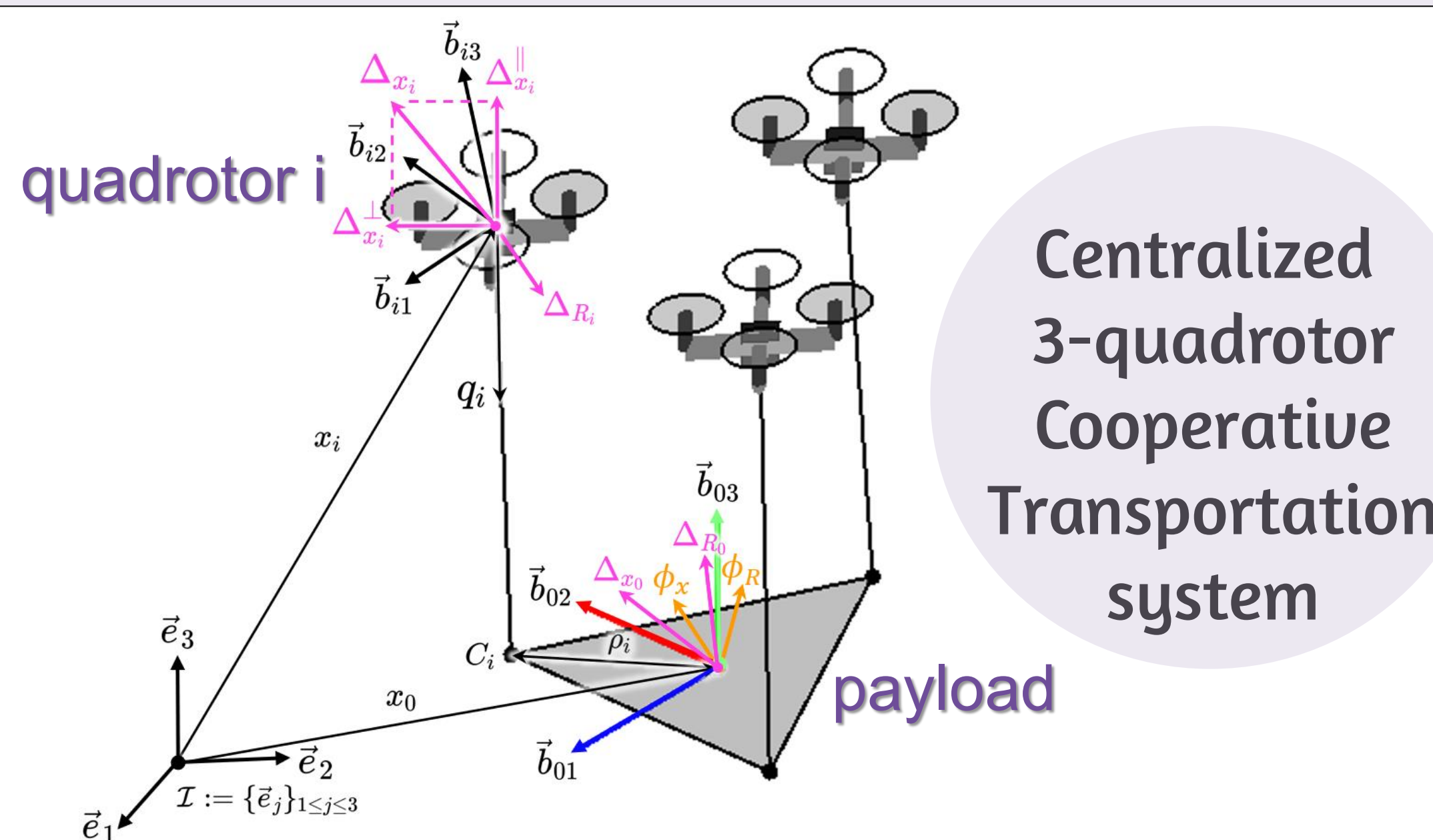
Video



ABSTRACT

This paper introduces an **adaptive-neuro geometric control** for a centralized **multi-quadrotor cooperative transportation system**, which enhances both adaptivity and disturbance rejection. Our strategy is to coactively tune the model parameters and learn the external disturbances in real-time. To realize this, we augmented the existing geometric control with multiple **neural networks** and **adaptive laws**, where the estimated model parameters and the weights of the neural networks are simultaneously tuned and adjusted **online**. The Lyapunov-based adaptation guarantees bounded estimation errors without requiring either pre-training or the persistent excitation (PE) condition. The proposed control system has been proven to be stable in the sense of Lyapunov under certain preconditions, and its enhanced robustness under scenarios of disturbed environment and model-unmatched plant was demonstrated by numerical simulations.

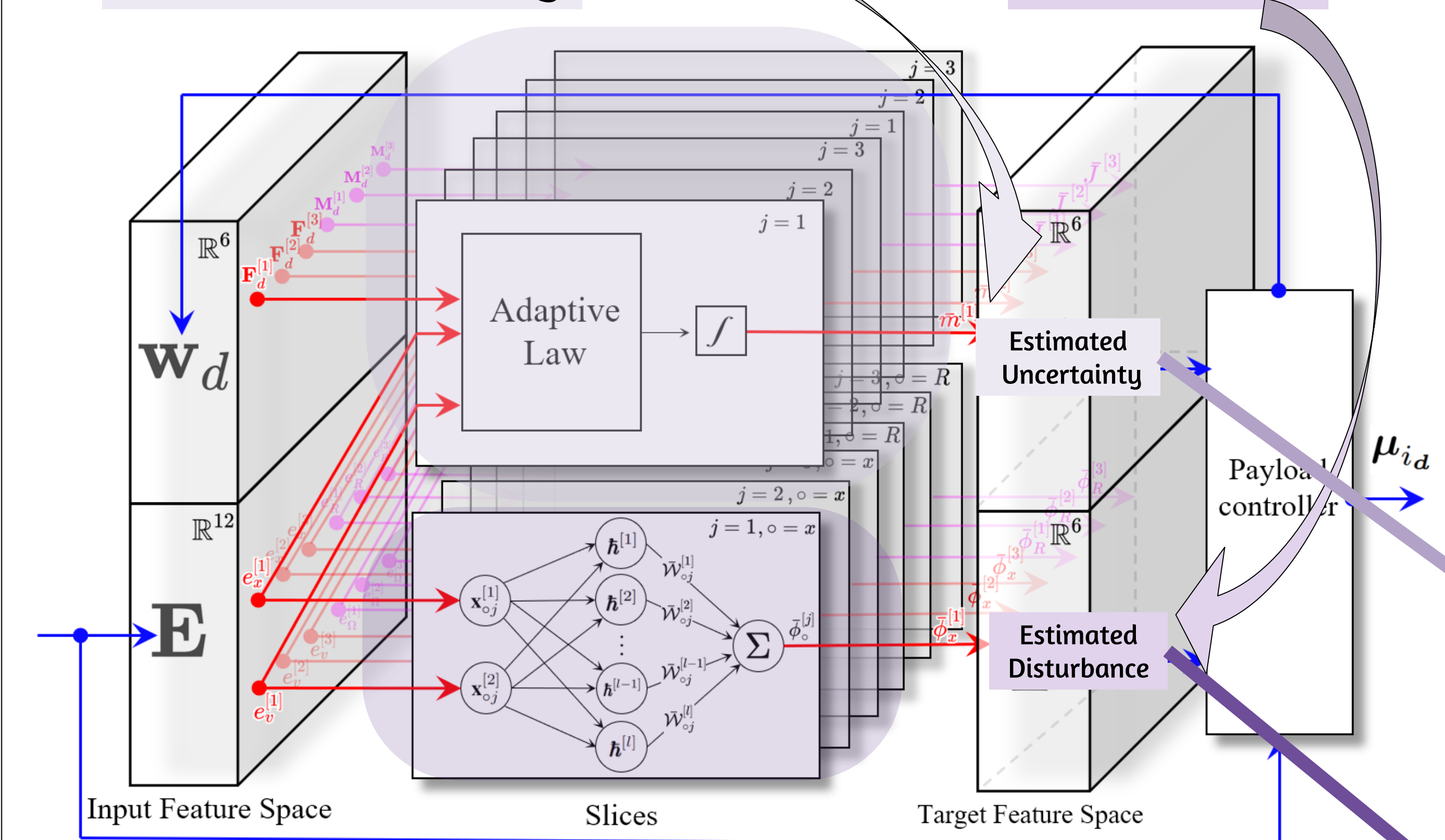
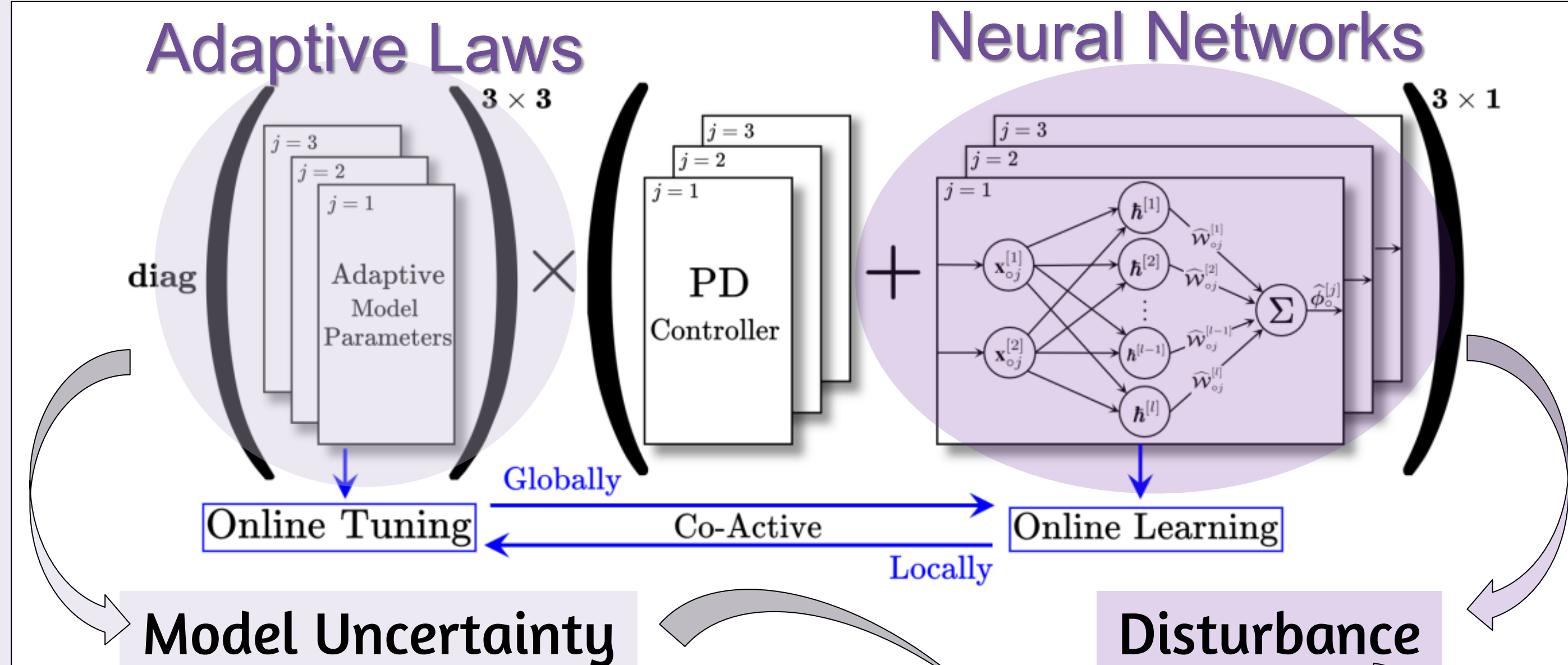
MODELING



Dynamics

$$\begin{cases} \dot{\Omega}_i = J_i^{-1}(\mathbf{M}_i - \Omega_i \times J_i \Omega_i + \Delta \mathbf{R}_i), \\ \dot{\mathbf{R}}_i = \mathbf{R}_i[\Omega_i]_{\times}, \\ \ddot{\mathbf{q}}_i = \frac{1}{m_i l_i} [q_i]_{\times}^2 (\mathbf{u}_i + \Delta \mathbf{x}_i - m_i \mathbf{a}_i) - \|\dot{\mathbf{q}}_i\|_2^2 \mathbf{q}_i, \\ \dot{\omega}_i = \frac{1}{l_i} [q_i]_{\times} \mathbf{a}_i - \frac{1}{m_i l_i} [q_i]_{\times} (\mathbf{u}_i^{\perp} + \Delta \mathbf{x}_i^{\perp}), \\ \ddot{\mathbf{x}}_0 = \frac{1}{m_0} \left(\mathbf{F}_d + \Delta \mathbf{x}_0 + \sum_{i=1}^n \Delta \mathbf{x}_i \right) - g \mathbf{e}_3 + \mathbf{Y}_x + \phi_x, \\ \dot{\Omega}_0 = J_0^{-1} \left(\mathbf{M}_d - [\Omega_0]_{\times} J_0 \Omega_0 + \Delta \mathbf{R}_0 + \sum_{i=1}^n [\rho_i]_{\times} \mathbf{R}_0^{\top} \Delta \mathbf{x}_i \right) \\ + \mathbf{Y}_R + \phi_R, \\ \dot{\mathbf{R}}_0 = \mathbf{R}_0[\Omega_0]_{\times}, \end{cases}$$

PROPOSED APPROACH



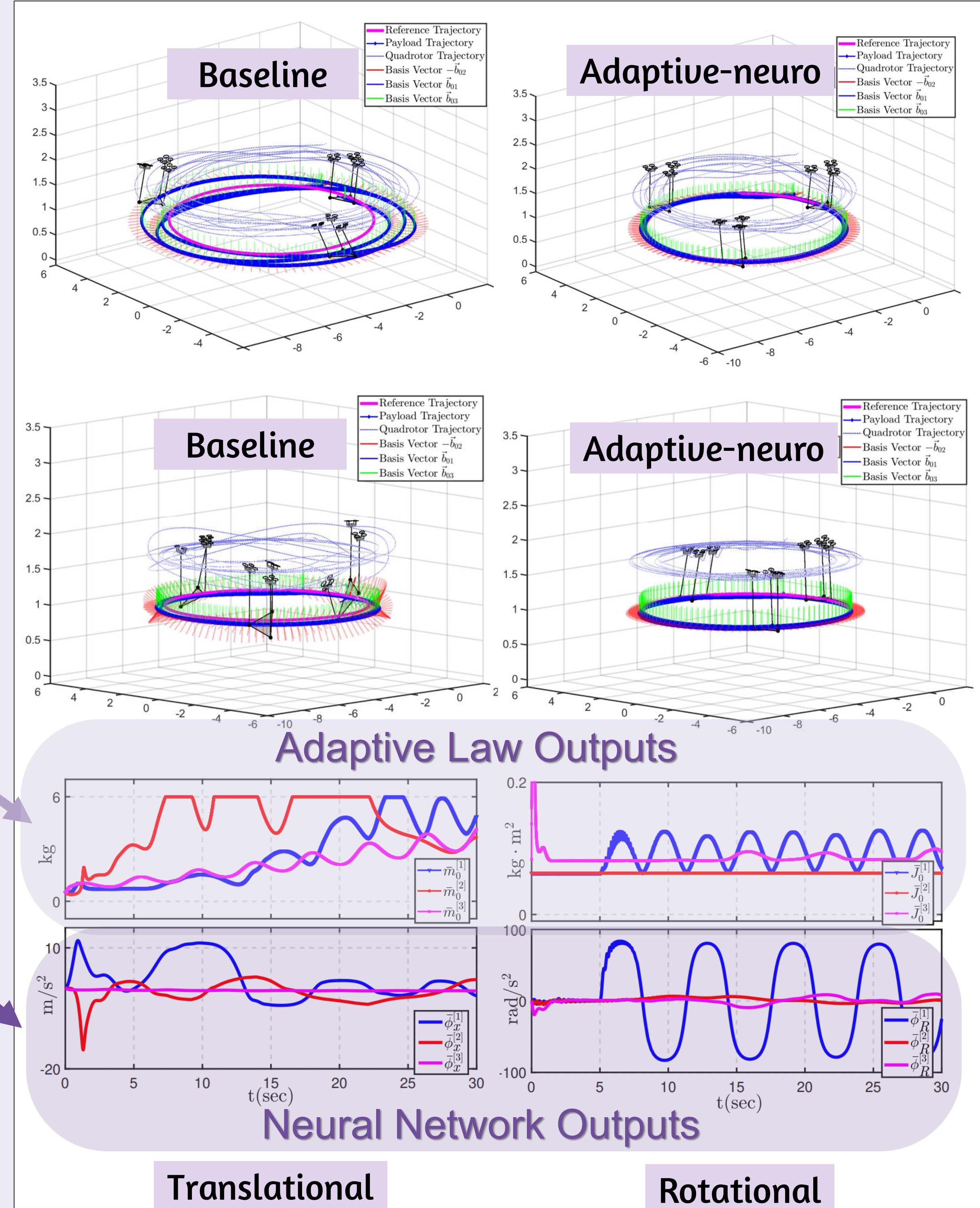
Input
(Errors)

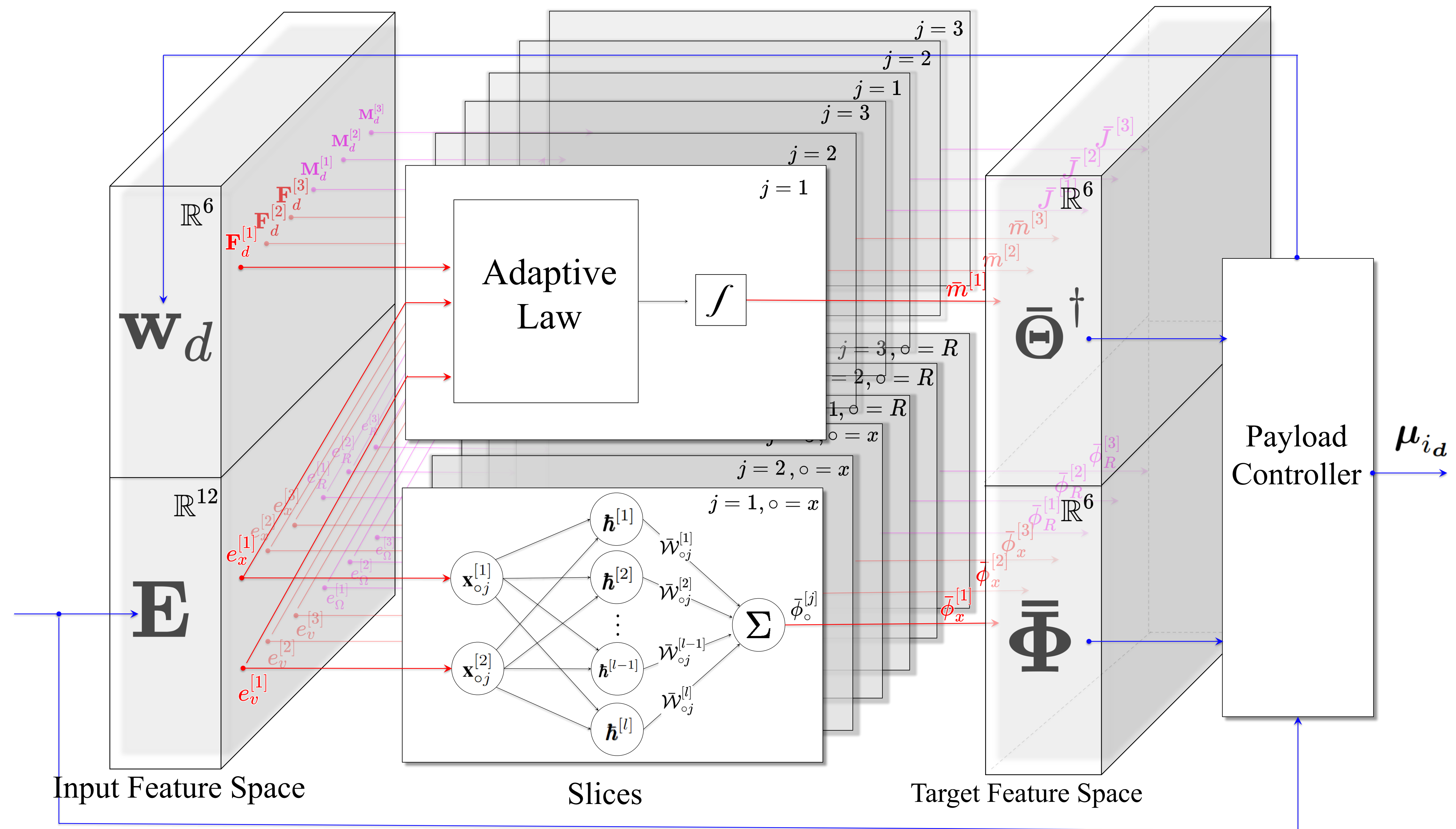
Submappings
(Learning)

Output
(Estimated)

Learning and tuning on **subspaces**
(A dimension decomposition Strategy)

SIMULATION





190%